

Excel Dashboard Build Guide

File Structure Reference

Raw Data tab:

- Row 1: Headers (Transaction_ID, Year, Quarter, Product, Revenue, Cost, Profit)
- Rows 2-187: Transaction data (186 rows)
- Year column: Numeric (2023, 2024)
- Quarter column: Text (Q1, Q2, Q3, Q4)

Dashboard tab:

- Row 7: Column headers (Product, Quarter, Total Revenue, Weighted Avg Margin, Rolling 3-Mo Trend, YoY Margin Delta, Margin Health)
 - Rows 8-39: Data rows (32 rows = 4 products × 8 quarters)
 - Column A: Product name
 - Column B: Quarter as "2023 Q1" format
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MILESTONE 1: Calculate Revenue & Weighted Margin

Step 1: Total Revenue (Column C)

Cell C8, enter:

None

```
=SUMIFS('Raw Data'!$E$2:$E$187, 'Raw Data'!$D$2:$D$187, $A8, 'Raw Data'!$B$2:$B$187, VALUE(LEFT($B8,4)), 'Raw Data'!$C$2:$C$187, RIGHT($B8,2))
```

This formula:

- Sums Revenue (column E) from Raw Data
- Matches Product (column D) to cell A8
- Matches Year (column B) to first 4 characters of B8, converted to a number
- Matches Quarter (column C) to last 2 characters of B8 (e.g., "Q1")

Format: Select C8 → Right-click → Format Cells → Currency, 0 decimals

Copy: Select C8 → Copy (Cmd+C) → Select C9:C39 → Paste (Cmd+V)

Step 2: Weighted Average Margin (Column D)

Cell D8, enter:

None

```
=SUMPRODUCT('Raw Data'!$D$2:$D$187=$A8)*('Raw Data'!$B$2:$B$187=VALUE(LEFT($B8,4)))*('Raw Data'!$C$2:$C$187=RIGHT($B8,2))*'Raw Data'!$G$2:$G$187)/C8
```

This formula:

- Sums Profit (column G) only for matching Product/Year/Quarter
- Divides by Total Revenue to get profit margin
- Automatically weights by revenue (larger transactions contribute more)

Format: Select D8 → Right-click → Format Cells → Percentage, 1 decimal

Copy: Select D8 → Copy → Select D9:D39 → Paste

✓ **Checkpoint:** Pick any product/quarter, manually sum its Profit ÷ Revenue from Raw Data. Should match column D.

MILESTONE 2: Add Trend & Comparison Columns

Step 3: Rolling 3-Month Trend (Column E)

This column shows the change in margin from the previous quarter. The first quarter of each product (2023 Q1) shows "N/A" since there's no prior quarter to compare.

Cell E8, enter:

None

```
=IF(AND(LEFT($B8,4)="2023",RIGHT($B8,2)="Q1"),"N/A",D8-D7)
```

This formula:

- Shows "N/A" for 2023 Q1 rows (rows 8, 16, 24, 32 — first data point for each product)
- Otherwise calculates current margin minus previous row's margin

Format: Select E8 → Right-click → Format Cells → Percentage, 1 decimal

Copy: Select E8 → Copy → Select E9:E39 → Paste

Step 4: Year-over-Year Margin Delta (Column F)

This column compares margin to the same quarter last year. All 2023 quarters show "N/A" since there's no 2022 data.

Cell F8, enter:

None

```
=IF(LEFT($B8,4)="2023", "N/A", D8-INDEX($D$8:$D$39, MATCH($A8&LEFT($B8,4)-1&RIGHT($B8,2), $A$8:$A$39&LEFT($B$8:$B$39,4)&RIGHT($B$8:$B$39,2), 0)))
```

This formula:

- Returns "N/A" for all 2023 rows
- For 2024 rows: finds the same product + same quarter from 2023 and calculates the difference

Note: This is an array-style MATCH. In older Excel versions, you may need to press **Ctrl+Shift+Enter** instead of just Enter.

Format: Select F8 → Right-click → Format Cells → Percentage, 1 decimal

Copy: Select F8 → Copy → Select F9:F39 → Paste

Step 5: Margin Health Classification (Column G)

Cell G8, enter:

None

```
=IF(D8>0.35, "Strong", IF(D8>=0.2, "Moderate", "At Risk"))
```

This formula:

- "Strong" if margin > 35%
- "Moderate" if margin is 20% to 35%
- "At Risk" if margin < 20%

Copy: Select G8 → Copy → Select G9:G39 → Paste

Optional Formatting: Select G8:G39 → Home → Conditional Formatting → New Rule:

- Text equals "Strong" → Green fill
- Text equals "Moderate" → Yellow fill
- Text equals "At Risk" → Red fill

✓ **Checkpoint:** Verify classifications match the margin values in column D.

MILESTONE 3: Create the Combo Chart

Step 6: Set Up Chart Data Table

The main summary table isn't structured for easy charting (products repeat in rows). Create a pivot-style table for the chart.

Cell A42, type: **Chart Data**

Row 43 - Headers:

Cell	Enter
------	-------

A43	Quarter
-----	---------

B43	Widget Pro
-----	------------

C43	Widget Standard
-----	--------------------

D43	Service Package
-----	--------------------

E43 Accessory Kit

F43 Total Revenue

Column A - Quarters (A44:A51):

Cell Enter

A44 2023 Q1

A45 2023 Q2

A46 2023 Q3

A47 2023 Q4

A48 2024 Q1

A49 2024 Q2

A50 2024 Q3

A51 2024 Q4

Step 7: Add Chart Data Formulas

These formulas pull margin data from your summary table above.

Cell B44 (Widget Pro margin for 2023 Q1):

None

```
=SUMPRODUCT(( $A$8:$A$39="Widget  
Pro")*($B$8:$B$39=$A44)*($D$8:$D$39))
```

Cell C44 (Widget Standard):

None

```
=SUMPRODUCT(( $A$8:$A$39="Widget  
Standard")*($B$8:$B$39=$A44)*($D$8:$D$39))
```

Cell D44 (Service Package):

None

```
=SUMPRODUCT( ( $A$8:$A$39="Service  
Package" )*( $B$8:$B$39=$A44)*( $D$8:$D$39) )
```

Cell E44 (Accessory Kit):

None

```
=SUMPRODUCT( ( $A$8:$A$39="Accessory  
Kit" )*( $B$8:$B$39=$A44)*( $D$8:$D$39) )
```

Cell F44 (Total Revenue for the quarter):

None

```
=SUMIF( $B$8:$B$39, $A44, $C$8:$C$39)
```

Copy formulas down:

- Select B44:F44 → Copy → Select B45:F51 → Paste

Format:

- Select B44:E51 → Format as Percentage (1 decimal)
- Select F44:F51 → Format as Currency (0 decimals)

Step 8: Create the Chart

1. **Select data:** Click and drag to select A43:F51 (headers + all data)
2. **Insert chart:**
 - Go to **Insert** tab → **Column** → **Clustered Column**
3. **Move Total Revenue to secondary axis:**
 - Click once on the chart
 - Click on any Total Revenue bar (the tallest bars) to select that series
 - Right-click → **Format Data Series**
 - Under Series Options, select **Secondary Axis**

4. **Change Total Revenue to a line:**

- With Total Revenue still selected
- Go to **Chart Design** tab → **Change Chart Type**
- Find Total Revenue in the list, change to **Line**
- Make sure Secondary Axis is checked
- Click **OK**

5. **Add chart title:**

- Click on the chart
- **Chart Design** → **Add Chart Element** → **Chart Title** → **Above Chart**
- Type: **Quarterly Margin Trends by Product**

6. **Add axis titles:**

- **Chart Design** → **Add Chart Element** → **Axis Titles** → **Primary Vertical**
- Type: **Profit Margin**
- **Chart Design** → **Add Chart Element** → **Axis Titles** → **Secondary Vertical**
- Type: **Total Revenue (\$)**

7. **Position and resize:**

- Drag the chart below your data tables
- Resize by dragging corners as needed